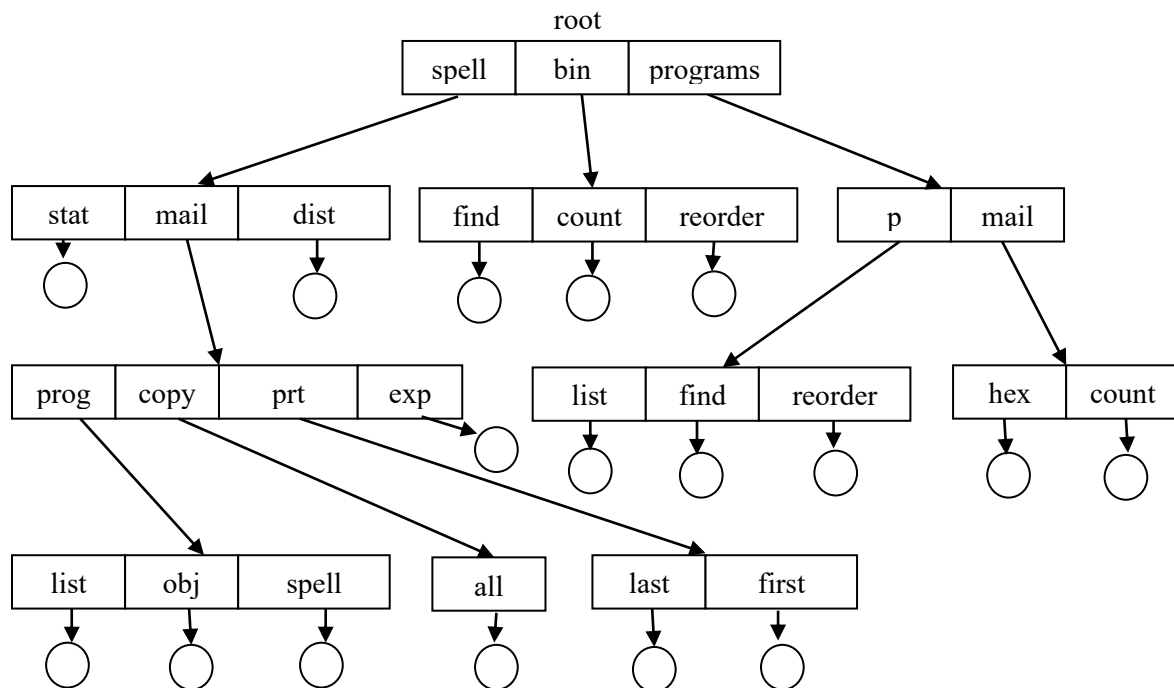


Chapter 9: File Management

ANSWERS

1. Consider the following directory structure:



Nyatakan (*Specify*):

- a. nama laluan mutlak bagi merujuk fail 'all'.
absolute path name to refer the file 'all'.

spell/mail/copy/all

- b. nama laluan relatif bagi merujuk fail 'all', jika direktori semasa ialah 'mail'.
relative path name to refer the file 'all', if the current directory is 'mail'.

copy/all

2. Assuming there are 16 blocks on a disk. Given the table below, show the storage location of **assign1.asm** file using the indexed allocation with variable length portions method.

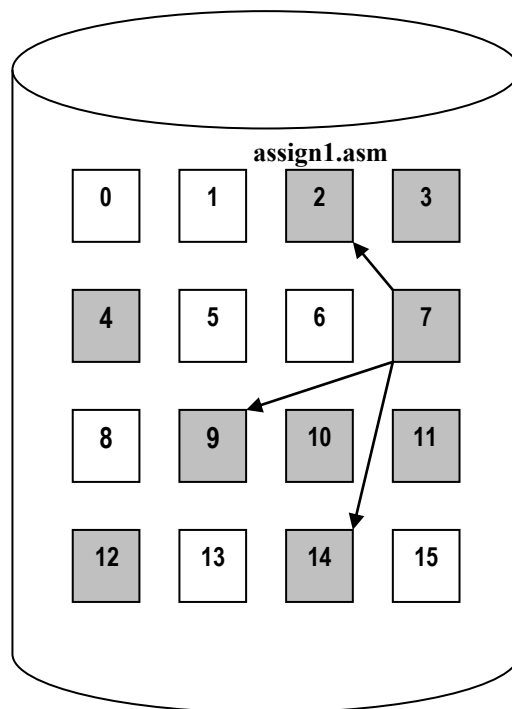
File Allocation Table

File Name	Index Block
assign1.asm	7

Start Block	Length
2	3
14	1
9	4

Use pointers (arrows) and shade the blocks involved.

File Allocation



3. Give two objectives of the file management system.

Any 2 of these answers:

- Meet the data management needs and requirements of the user
- Guarantee that the data in the file are valid
- Optimize performance
- Provide I/O support for a variety of storage device types
- Minimize or eliminate the potential for lost or destroyed data
- Provide a standardized set of I/O interface routines
- Provide I/O support for multiple users

4. Give two types of file organizations and explain each of them.

- The File
 - Data are collected in the order they arrive
 - Purpose is to accumulate a mass of data and save it
 - Records may have different fields
 - No structure
 - Record access is by exhaustive search
- The Sequential File
 - Fixed format used for records
 - Records are the same length
 - All fields the same (order and length)
 - Field names and lengths are attributes of the file
 - One field is the key field
 - Uniquely identifies the record
 - Records are stored in key sequence
- Indexed Sequential File
 - Index provides a lookup capability to quickly reach the vicinity of the desired record
 - Contains key field and a pointer to the main file
 - Indexed is searched to find highest key value that is equal to or precedes the desired key value
 - Search continues in the main file at the location indicated by the pointer
- Indexed File
 - Uses multiple indexes for different key fields
 - May contain an exhaustive index that contains one entry for every record in the main file
 - May contain a partial index
- The Direct or Hashed File
 - Directly access a block at a known address
 - Key field required for each record

5. List and explain SIX (6) basic file operations.

- Create - define new file and position it within file structure.
- Delete - remove from the file structure and destroyed.
- Open - to allow a process to perform functions on it.
- Close - close with respect to a process.
- Read - read all or a portion of a file.
- Write (update) - add new data, or change values.

6. Give and explain TWO (2) file access methods. How do they differ from each other?
- i. Sequential access - Information in the file is processed in order
 - ii. Direct access - Disk allow random access to any file block

7. File organization is the logical structuring of records as how they are accessed. Files are organized to suit certain criteria. Give five (5) criteria for file organization.

- Short access time
 - Needed when accessing a single record
 - Not needed for batch mode
- Ease of update
 - File on CD-ROM will not be updated, so this is not a concern
- Economy of storage
 - Should be minimum redundancy in the data
 - Redundancy can be used to speed access such as an index
- Simple maintenance
- Reliability

8. There are three methods of record blocking. Briefly explain each of them.

- Fixed blocking
 - Fixed length records.
 - An integral number of records are stored in a block
 - Possible internal fragmentation.
- Variable-length spanned blocking
 - Variable length records, no limit to record size.
 - Packed into blocks with no unused space - some records must span two blocks, with the continuation indicated by a pointer.
 - (-) Records spanning 2 blocks require 2 I/O operations.
- Variable-length unspanned blocking
 - Variable length records.
 - No spanning employed - limits record size $\leq$ block size.
 - Possible internal fragmentation.